

An Adaptive Grouped-Heterogeneity Estimator for Cross-Sectional Data

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Setup

Consider the cross-sectional regression model

$$Y_i = \beta(W_i)'X_i + r_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid X_i, W_i] = 0$$

Researcher observes (Y_i, X_i, W_i) for $i = 1, \dots, n$

- Covariate $X_i \in \mathbb{R}^d$, possibly continuous;
- Discrete covariate $W_i \in \{1, \dots, J\}$ with potentially large support, $J \rightarrow \infty$;
- Approximation error r_i vanishes as the sample size grows.

Write $\beta_j := \beta(j)$. Object of interest is the full collection $(\beta_1, \dots, \beta_J)$.

Naive Estimator

Treat values of W_i as unrelated and estimate a separate coefficient in each cell;

$$\hat{\beta}_j := \arg \min_{b \in \mathbb{R}^d} \sum_{i=1}^n (Y_i - b' X_i)^2 \mathbb{1}\{W_i = j\}$$

Problem: Precision is governed by the number of observations in a cell, $n_j \approx n/\mathcal{J}$, rather than by the sample size.

- Object of interest is the full collection, so error accumulates across cells,

$$\sum_{j=1}^{\mathcal{J}} \|\hat{\beta}_j - \beta_j\|^2 \asymp_p \frac{d\mathcal{J}^2}{n};$$

- When $\mathcal{J}$ is large relative to n , this converges slowly or diverges.

Nonparametric Regression

Leading example is nonparametric estimation of $\mathbb{E}[Y_i \mid \tilde{W}_i, V_i]$, where $\tilde{W}_i$ collects several discrete covariates and V_i is an optional continuous covariate.

Let W_i index the saturated cells generated by $\tilde{W}_i$. A series representation within each cell, with $p_d(V_i)$ a vector of d basis functions of V_i , gives

$$Y_i = \beta(W_i)'X_i + r_i + \varepsilon_i, \quad X_i = p_d(V_i) \in \mathbb{R}^d$$

- Without continuous covariates, $X_i = 1$, $r_i = 0$, and $\beta(w) = \mathbb{E}[Y_i \mid W_i = w]$;
- Approximation error can be made small by increasing d with the sample size (Stone, 1985, Chen, 2007).

Number of cells grows exponentially in the number of covariates: 10 binary covariates give $J = 2^{10} = 1,024$.

Nonparametric Regression

Blandhol et al. (2026) show that the LATE interpretation of 2SLS requires a nonparametric first stage, which with discrete instruments or controls means saturating.

- Saturated first stage is exactly the cell-by-cell regression above;
- Their three reanalyses, the first of which is Card (1995), have $(J, n/J) \approx (14, 220)$, $(99, 170)$, and $(66, 50)$.

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- Saturated first stage is exactly the cell-by-cell regression above;
- Their three reanalyses, the first of which is [Card \(1995\)](#), have $(\mathcal{J}, n/\mathcal{J}) \approx (14, 220)$, $(99, 170)$, and $(66, 50)$.

Interacting the original controls in that first application gives

$$\mathcal{J} = 819, \quad n/\mathcal{J} \approx 4$$

with 358 cells containing a single observation. Our application coarsens this to $\mathcal{J} = 92$, where $n/\mathcal{J} \approx 33$.

Group Structure

Naive estimator does not take advantage of similarities between cells. Among the 819 cells in Card (1995), it might be reasonable to assume that many share a first-stage coefficient.

Formally, suppose the collection $\{\beta_1, \dots, \beta_J\}$ takes at most G distinct values. Cells sharing a value form the groups

$$g(j) := \{k \in [J] : \beta_k = \beta_j\}$$

Neither the number of distinct vectors, G , nor the partition is known to the researcher.

Goal: Want an estimator that

1. Adapts to group structure when present;
2. Does no worse than the naive estimator when it is absent.

Examples: Judge Leniency

In Dahl et al. (2014), disability-insurance applicants are randomly assigned to appeal judges and judge leniency instruments for allowance.

- W_i indexes the judge assigned to case i , $X_i = 1$, and $\beta(w)$ is judge w 's leniency;
- $J = 79$ judges and $n/J \approx 190$.

Judges may share evidentiary thresholds, readings of the disability criteria, or institutional backgrounds.

These latent types are not recorded in the assignment data.

Examples: Many Fixed Effects

Same structure appears in many other settings. Simplest case is

$$Y_i = \beta(W_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid W_i] = 0$$

with $X_i = 1$; more generally Y_i is residualized.

- Teacher value-added (Kane et al., 2008): $J = 10,040$ teachers, $n/J \approx 150$;
- Medicare Advantage (Abaluck et al., 2021): $J = 75,417$ plan products, $n/J \approx 200$;
- Firm-level hiring (Kline et al., 2022): $J = 108$ firms, $n/J \approx 770$.

Unit label may be finer than the economically relevant heterogeneity.

- Teachers share training programs;
- Plans belong to insurer networks;
- Firms use common hiring practices.

Examples: Geographic Heterogeneity

Effects allowed to vary across regions. After partialling out common controls,

$$\tilde{Y}_i = \beta(W_i)' \tilde{X}_i + \varepsilon_i, \quad W_i = \text{region}$$

A large national sample does not give precision for every region, since each $\beta(w)$ uses within-region variation only.

- Chetty et al. (2013): EITC knowledge across $J = 899$ three-digit zip areas;
- Chetty and Hendren (2018): childhood exposure effects, $J = 741$ commuting zones or $J = 3,138$ counties.

Economically similar regions need not be adjacent, so the relevant partition may not be apparent from the region labels.

Examples: Occupational Wage Premia

Log wages allowed to vary across detailed occupation codes, with the occupation effect interpreted as the price of occupation-specific skills.

For occupation-specific wage growth, set $X_i = (1, \text{Post}_i)'$.

- First coefficient is the baseline premium;
- Second is the occupation-specific change.

Autor and Dorn (2013) study these changes from 1980 to 2005 for $J = 318$ detailed nonfarm occupations.

Occupation codes need not capture the economic sources of wage growth.

Alternatives: Machine Learning

ML methods typically applied in high-dimensional settings may not be well suited for applications with discrete covariates.

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None of these methods takes advantage of similarities between cells.

Alternatives: Clustering

Cluster $\hat{\beta}_1, \dots, \hat{\beta}_J$ into G groups and pool within cluster.

1. **Choosing G .** Requires knowing or consistently estimating the number of groups. Absent separation there is no consistent rule, and cross-validating over G is itself unstable when there is no partition to find.

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3. **Computation.** k -means is NP-hard as d grows (Dasgupta, 2008, Aloise et al., 2009) and, in fixed dimension, as the number of clusters grows (Mahajan et al., 2012).

Adapts when groups are strong, but can do worse than the naive estimator when they are weak or absent.

Alternatives: Empirical Bayes

Empirical Bayes models the distribution of effects (Robbins, 1956, James and Stein, 1961, Armstrong et al., 2022, Chen, 2026); see Walters (2024) for an overview. In the scalar case,

$$\hat{\beta}_j \mid \beta_j \sim N(\beta_j, \sigma^2), \quad \beta_j \sim F, \quad \hat{\beta}_j^{\text{EB}} = \mathbb{E}_{\hat{F}}[\beta_j \mid \hat{\beta}_j]$$

- Empirical Bayes rules are separable given the estimated prior: cell j 's estimate depends on other cells only through $\hat{F}$. Our estimator pools the estimates directly.
- Gaussian $\hat{F}$ (Kane et al., 2008, Abaluck et al., 2021) smooths across groups; nonparametric $\hat{F}$ does adapt to discrete priors (Jiang and Zhang, 2009, Saha and Guntuboyina, 2020).
- That adaptivity is a regret bound against the Bayes rule for the true prior, in a homoskedastic sequence model with known σ .

To our knowledge, no empirical Bayes procedure has been shown to attain the rate of an oracle that knows the group structure.

Preview of Estimator

Start from the naive cell-by-cell estimates.

1. **Pool.** Exponentially weighted average, with larger weight on cells that are close in a variance-adjusted metric. No hard assignment to groups;
2. **Debias.** Correct the bias from using $\hat{\beta}_j$ in its own weights;
3. **Recombine.** Convex combination with the naive estimator, with weight chosen by SURE; the amount of pooling retained is determined by the data.

Requires one deterministic weighting calculation, and no partition is estimated.

Pooling delivers adaptivity; recombination provides protection when groups are weak or absent.

Preview of Results

1. Can the gains from pooling be obtained without knowing the number of groups or the assignment of cells to groups?
 - Answer: Yes. Under a fixed number of well-separated groups, the estimator attains the aggregate rate of an oracle that knows the partition, up to logarithmic factors.
2. Group structure is generally not verifiable. Is anything lost when it is weak or absent?
 - Answer: No. Aggregate quadratic loss is asymptotically no larger than that of the naive estimator, with no separation or grouping condition imposed.

Both established first in a Gaussian sequence model, then with estimated covariances, non-Gaussian errors, growing d , and nonzero approximation error.

Prior Literature

1. **Unobserved Group Structure in Panel Data.** Lin and Ng (2012), Bonhomme and Manresa (2015), Su et al. (2016), Ando and Bai (2016), Bonhomme et al. (2022), Yu et al. (2024), Chetverikov and Manresa (forthcoming).
 - Our contribution: Consider a cross-sectional setting with an observed discrete covariate and no variation over time from which to learn the groups.
2. **Empirical Bayes and Many Noisy Parameters.** Robbins (1956), James and Stein (1961), Kane et al. (2008), Abaluck et al. (2021), Armstrong et al. (2022), Gu and Koenker (2023), Walters (2024), Chen (2026), Song et al. (2026); nonparametric maximum likelihood and regret bounds in Jiang and Zhang (2009), Saha and Guntuboyina (2020), Polyanskiy and Wu (2021), Soloff et al. (2025), Zhang and Volgushev (2026).
 - Our contribution: Regularize by equality of coefficients across cells rather than by estimating a distribution of effects.
3. **Bilateral Filtering.** Tomasi and Manduchi (1998), Durand and Dorsey (2002), Paris and Durand (2006), Paris et al. (2009).
 - Our contribution: Retain the same averaging principle, adding debiasing and SURE recombination to obtain large-sample guarantees.

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Sequence Model

Suppose the researcher directly observes preliminary estimates

$$\hat{\beta}_j = \beta_j + \sigma \xi_j, \quad \xi_j \sim N(0, \Sigma_j), \quad j = 1, \dots, J$$

independently across j , with

- Σ_j positive definite, known, possibly different across cells;
- Common noise scale $\sigma^2 = J/n \rightarrow 0$;
- Fixed d and zero approximation error.

Standard motivation is a within-cell central limit theorem: with order n/J observations per cell, least squares is approximately Gaussian with covariance of order J/n .

Naive estimator is $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_J)$.

Benchmarks

Work with aggregate quadratic loss $\sum_{j=1}^J \|\hat{\beta}_j - \beta_j\|^2$.

Naive.

$$\sum_{j=1}^J \|\hat{\beta}_j - \beta_j\|^2 \asymp_p \frac{J^2}{n}$$

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Naive.

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Group oracle. If the partition were known, take the precision-weighted average within each group,

$$\hat{\beta}_j^{\text{or}} := \left(\sum_{k \in g(j)} \Sigma_k^{-1} \right)^{-1} \sum_{k \in g(j)} \Sigma_k^{-1} \hat{\beta}_k, \quad \mathbb{E} \sum_{j=1}^J \|\hat{\beta}_j^{\text{or}} - \beta_j\|^2 \asymp \frac{J}{n}$$

Pooling over $\asymp J$ cells per group improves the rate by a factor of J .

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Pooling over $\asymp J$ cells per group improves the rate by a factor of J .

Key question: Can this be obtained without knowing the partition, and without doing worse than the naive estimator when groups are weak or absent?

Step 1: Exponentially Weighted Estimator

Weight preliminary estimates by proximity in a variance-adjusted metric;

$$\omega_{jk} := \frac{\exp\left(-\frac{1}{2\sigma^2} \|\hat{\beta}_j - \hat{\beta}_k\|_{\Omega_k}^2\right)}{\sum_{\ell=1}^J \exp\left(-\frac{1}{2\sigma^2} \|\hat{\beta}_j - \hat{\beta}_\ell\|_{\Omega_\ell}^2\right)}, \quad \Omega_\ell := \gamma(I_d - \gamma\Sigma_\ell)^{-1}$$

where $\|v\|_A^2 = v'Av$ and $\gamma > 0$ is a tuning parameter.

- Ω_ℓ is increasing in Σ_ℓ , so ω_{jk} is largest when $\hat{\beta}_k$ is close to $\hat{\beta}_j$ and precisely estimated. This form is chosen so that the shift removed in Step 2 has a closed form.
- Small γ gives $\omega_{jk} \approx \mathcal{I}^{-1}$, the simple mean; as γ grows, the weights concentrate on the cells closest to j .

Note: For fixed d , γ does not shrink with the sample size. Our unified default is $\gamma = 0.2/[d \max_\ell \lambda_{\max}(\Sigma_\ell)]$.

Step 1: Exponentially Weighted Estimator

Take the scalar case $\Sigma_j = 1$, and write

$$\hat{\beta}_j = \beta_j + \sigma Z_j, \quad Z_j \sim N(0, 1), \quad \Omega = \frac{\gamma}{1 - \gamma}$$

Then

$$\omega_{jk} \propto \begin{cases} \exp(-\Omega(Z_j - Z_k)^2/2), & k \in g(j) \\ \exp(-\Omega[\delta_{jk}/\sigma + O_p(1)]^2/2), & k \notin g(j) \end{cases}$$

where $\delta_{jk} = |\beta_j - \beta_k|$ is bounded away from zero under the separation condition stated below.

- Since $\sigma \rightarrow 0$, ω_{jk} for $k \notin g(j)$ quickly tends to zero;
- Within $g(j)$, weights remain random as $n, J \rightarrow \infty$.

Step 1: Exponentially Weighted Estimator

Pool the preliminary estimates with these weights;

$$\check{\beta}_j := \sum_{k=1}^J \omega_{jk} \hat{\beta}_k$$

- Computable from $\hat{\beta}_1, \dots, \hat{\beta}_J$ and the known Σ_j , with no estimate of the partition;
- As $n, J \rightarrow \infty$, weight outside $g(j)$ vanishes and $\check{\beta}_j$ is an average of $\asymp J$ estimates of the same β_j .

Averaging $\asymp J$ independent, mean β_j estimates, would cut the error from σ to $\sigma/\sqrt{J} \asymp n^{-1/2}$, the oracle rate.

Weights within $g(j)$ remain random. To see what this does to the average, condition on $\hat{\beta}_j$.

Step 2: Debiasing

Continue the scalar case and condition on $\hat{\beta}_j$, equivalently on Z_j . Within $g(j)$,

$$\check{\beta}_j \approx \sum_{k \in g(j)} \tilde{\omega}_{jk} \hat{\beta}_k, \quad \tilde{\omega}_{jk} := \frac{r_{jk}}{\sum_{\ell \in g(j)} r_{j\ell}}, \quad r_{jk} := e^{-\Omega(Z_j - Z_k)^2/2}$$

- Given Z_j , the denominator is a sum of $\asymp \mathcal{J}$ independent terms and can be treated as fixed;
- Numerators $r_{jk} \hat{\beta}_k$ are independent across k .

$\check{\beta}_j$ is then a sample mean of independent terms and converges to its conditional mean.

Problem: $\hat{\beta}_k$ is independent of $\hat{\beta}_j$ and its conditional mean is still β_j . The weight r_{jk} is not independent: it is largest when Z_k is near Z_j , so the weighted average of noise converges, given Z_j , to

$$\frac{\mathbb{E}[r_{jk} Z_k \mid Z_j]}{\mathbb{E}[r_{jk} \mid Z_j]} \neq 0$$

Conditional mean of $\check{\beta}_j$ is β_j plus σ times this limit. Gap is comparable to the naive error σZ_j and does not fall as cells are added to the average.

Step 2: Debiasing

Key Idea: Errors are Gaussian, so this conditional mean can be computed using properties of Gaussian integrals;

$$\frac{\mathbb{E}[r_{jk}Z_k \mid Z_j]}{\mathbb{E}[r_{jk} \mid Z_j]} = \gamma Z_j, \quad \mathbb{E}[\check{\beta}_j \mid \hat{\beta}_j] \approx (1 - \gamma)\beta_j + \gamma\hat{\beta}_j$$

Define the debiased estimator and substitute the conditional mean of $\check{\beta}_j$;

$$\tilde{\beta}_j := (1 - \gamma)^{-1}\check{\beta}_j - \gamma(1 - \gamma)^{-1}\hat{\beta}_j, \quad \mathbb{E}[\tilde{\beta}_j \mid \hat{\beta}_j] \approx \beta_j$$

Conditional on $\hat{\beta}_j$, $\tilde{\beta}_j$ is unbiased for β_j and approximately an average of $\asymp J$ independent terms, thus converges at oracle rate.

General case is the same calculation with $\Gamma_j := \gamma \sum_{k=1}^J \omega_{jk} \Sigma_k$ in place of γ ;

$$\tilde{\beta}_j := (I_d - \Gamma_j)^{-1}\check{\beta}_j - (I_d - \Gamma_j)^{-1}\Gamma_j\hat{\beta}_j$$

Naive estimator has error $O_p(\sqrt{J/n})$; the debiased estimator attains the oracle rate $O_p(n^{-1/2})$.

Step 3: Recombination

Reporting $\tilde{\beta}_j$ directly can do worse than the naive estimator in the absence of well-defined group structure.

- Without well-separated groups the weights can average across cells with different β_j ;
- Resulting bias need not be offset by the variance reduction, so $\sum_j \|\tilde{\beta}_j - \beta_j\|^2$ can exceed $\sum_j \|\hat{\beta}_j - \beta_j\|^2$.

Group structure is not verifiable, so recombine with the naive estimator;

$$\check{\beta}_j(\alpha) := \alpha \tilde{\beta}_j + (1 - \alpha) \hat{\beta}_j, \quad \alpha \in [0, 1]$$

- Infeasible target is $\mathcal{L}(\alpha) := \sum_{j=1}^J \|\check{\beta}_j(\alpha) - \beta_j\|^2$;
- Minimizing $\mathcal{L}$ would give an estimator no worse than the naive one, which is $\alpha = 0$.

Step 3: Recombination

Since $\mathcal{L}$ is infeasible, minimize Stein's unbiased risk estimate (Stein, 1981);

$$\hat{\mathcal{L}}(\alpha) := \sum_{j=1}^J \|\hat{\beta}_j - \check{\beta}_j(\alpha)\|^2 + 2\sigma^2 \sum_{j=1}^J \text{tr} \left(\Sigma_j \frac{\partial \check{\beta}_j(\alpha)}{\partial \hat{\beta}_j} \right) - \sigma^2 \sum_{j=1}^J \text{tr}(\Sigma_j),$$

$$\check{\alpha} := \arg \min_{\alpha \in [0,1]} \hat{\mathcal{L}}(\alpha)$$

- First term replaces the unknown β_j by $\hat{\beta}_j$; the trace term corrects for the correlation between $\check{\beta}_j(\alpha)$ and the noise in $\hat{\beta}_j$, and the constant removes the noise variance, so $\mathbb{E}\hat{\mathcal{L}}(\alpha) = \mathbb{E}\mathcal{L}(\alpha)$ at each fixed α ;
- Criterion is a convex quadratic in α , so $\check{\alpha}$ is the unconstrained minimizer projected onto $[0, 1]$;
- Jacobian of $\check{\beta}(\alpha)$ is $\alpha P_j + (1 - \alpha)I_d$, where $P_j := \partial \tilde{\beta}_j / \partial \hat{\beta}_j$ admits a closed form expression; no numerical differentiation is needed.

Final estimator is $\check{\beta}_j = \check{\alpha} \tilde{\beta}_j + (1 - \check{\alpha}) \hat{\beta}_j$, with a single $\check{\alpha}$ shared across all J cells.

Stein's Identity

Closed-Form Jacobian

Sequence Model: Conditions

Roughly, the results in this section require the following:

1. **Sequence model.** Preliminary estimates are independent and normal around the truth, with known covariances and a common noise scale that vanishes. Dimension is fixed.
2. **Comparable precision.** No cell is far more or less precisely estimated than the rest, though the covariances may still differ.
3. **Bounded tuning parameter.** γ small enough that the inverses in the estimator exist and the exponential moments are finite.
4. **Well-separated groups.** Distinct coefficient vectors are at least a fixed distance apart, their number is bounded as J grows, and each group holds a comparable share of the cells. Needed only for oracle performance under group-structure.

Items 1 to 3 form the sequence model condition and item 4 the separation condition.

Sequence Model

Separation

Sequence Model: Adaptivity

Theorem 1 (Group Adaptivity)

Suppose the sequence model $\mathfrak{J}$ and separation $\check{\mathfrak{J}}$ conditions hold, and $\mathfrak{J} \log^3(\mathfrak{J})/n \rightarrow 0$. Then the exponentially weighted estimator and the SURE weight satisfy

$$\sum_{j=1}^{\mathfrak{J}} \|\tilde{\beta}_j - \beta_j\|^2 = O_p\left(\frac{\mathfrak{J}}{n}\right), \quad 1 - \check{\alpha} = O_p\left(\frac{\log^{3/2} \mathfrak{J}}{\sqrt{\mathfrak{J}}}\right),$$

and the final estimator satisfies

$$\sum_{j=1}^{\mathfrak{J}} \|\check{\beta}_j - \beta_j\|^2 = O_p\left(\frac{\mathfrak{J} \log^3(\mathfrak{J})}{n}\right).$$

Interpretation:

- Exponentially weighted estimator attains the oracle order $\mathfrak{J}/n$ with no logarithmic loss;
- $\check{\alpha} \rightarrow 1$, so SURE puts asymptotically all weight on $\tilde{\beta}_j$; the logarithmic factor in the final bound comes from the vanishing weight left on the naive estimator.

Rates

	aggregate loss	average loss per cell
Naive	$\asymp_p \mathcal{J}^2/n$	$\asymp_p \mathcal{J}/n$
Group oracle	$\asymp \mathcal{J}/n$	$\asymp 1/n$
Exponentially weighted	$O_p(\mathcal{J}/n)$	$O_p(1/n)$
Final estimator	$O_p(\mathcal{J} \log^3(\mathcal{J})/n)$	$O_p(\log^3(\mathcal{J})/n)$

Bound improves on the naive rate by a factor $\mathcal{J}/\log^3(\mathcal{J})$.

Take $\mathcal{J} \asymp n^a$ with $1/2 < a < 1$. Naive aggregate loss diverges, while the oracle, exponentially weighted, and final losses all converge to zero.

Proof Strategy

Under group structure, the proof of Theorem 1 proceeds in three steps.

Proof Strategy.

1. **Outside** $g(j)$. Separation makes the weight on other groups exponentially small.
2. **Within** $g(j)$. By the logic above, conditional on $\hat{\beta}_j$, $\tilde{\beta}_j$ is approximately an unbiased weighted average of $\asymp \mathcal{J}$ independent terms, so $\tilde{\beta}_j$ converges at the oracle rate $O_p(n^{-1/2})$.
3. **Recombination**. SURE tracks the realized loss, so $\check{\alpha} \rightarrow 1$. Weight left on the naive estimator contributes

$$(1 - \check{\alpha})^2 O_p(\mathcal{J}^2/n) = O_p(\mathcal{J} \log^3(\mathcal{J})/n).$$

Next result makes Step 3 precise and imposes no group structure.

SURE Oracle Comparison

Write $\mathcal{L}(\alpha) := \sum_{j=1}^J \|\check{\beta}_j(\alpha) - \beta_j\|^2$ for the loss at mixing weight α , and define

$$R_J := \frac{J^{3/2} \log^{3/2}(J)}{n}$$

Proposition 1 (SURE Oracle Comparison)

Suppose the *sequence model* conditions hold. For every fixed $\alpha \in [0, 1]$, $\mathbb{E}\hat{\mathcal{L}}(\alpha) = \mathbb{E}\mathcal{L}(\alpha)$. Moreover,

$$\sup_{\alpha \in [0,1]} |\hat{\mathcal{L}}(\alpha) - \mathcal{L}(\alpha)| = O_p(R_J) \quad \text{and} \quad \mathcal{L}(\check{\alpha}) \leq \inf_{\alpha \in [0,1]} \mathcal{L}(\alpha) + O_p(R_J).$$

- Unbiasedness of SURE criterion follows from Stein's identity and alone does not justify selecting α via SURE. Use Gaussian Poincaré inequality to show that SURE is consistent.
- Second display is stronger than a comparison with the naive estimator. Up to $O_p(R_J)$, $\check{\alpha}$ attains the loss of the infeasible α that minimizes $\mathcal{L}$.

Sequence Model: Weak Domination

Separation is generally not verifiable, so the second result does not impose it.

Theorem 2 (Weak Domination of the Naive Estimator)

Suppose the sequence model $\mathfrak{B}$ conditions hold. Then

$$\sum_{j=1}^J \|\check{\beta}_j - \beta_j\|^2 \leq (1 + o_p(1)) \sum_{j=1}^J \|\hat{\beta}_j - \beta_j\|^2.$$

Discussion:

- No separation, finite-group, or equality restriction on the coefficients;
- Follows from Proposition 1 at the feasible benchmark $\alpha = 0$, which is the naive estimator.

Sequence Model: Discussion

1. Theorem 2 neither requires nor implies that $\check{\alpha} \rightarrow 0$ when exact groups are absent.
 - Coefficients that are close but unequal may still make partial pooling useful. Simulations illustrate this, though the theorem does not deliver it.
2. Similar in spirit to classical shrinkage results, where a misspecified Gaussian prior can still improve risk (James and Stein, 1961). Recent empirical Bayes work makes the same robustness argument (Armstrong et al., 2022, Walters, 2024).
 - Analogy is not exact. This is an asymptotic statement about the SURE combination rather than a finite-sample result.
3. Researcher does not need to know whether groups are present to benefit when they are.

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Relaxing the Model

Return to the regression model and relax the sequence experiment in four directions.

1. Covariance matrices unknown, replaced by a heteroskedasticity-robust estimate $\widehat{\Sigma}_j$;
2. Covariate dimension grows, $d \rightarrow \infty$;
3. Approximation error nonzero, $r_i \neq 0$;
4. Errors non-Gaussian, independent and sub-Gaussian with arbitrary heteroskedasticity.

Estimator itself is unchanged.

- Substitute $\widehat{\Sigma}_j$ in Ω_ℓ , Γ_j , and the SURE criterion;
- Covariates (X_i, W_i) are treated as fixed.

Extending the Argument

Preliminary estimates are approximately normal in large samples, but the extension is not immediate.

- Each weight ω_{jk} depends on all J preliminary estimates through the denominator;
- SURE criterion aggregates derivatives of those weights across every cell.

Two consequences:

- Central limit approximation applied to each $\hat{\beta}_j$ separately does not suffice;
- Gaussian approximation error must be tracked through each part of the estimator, then summed over $J \rightarrow \infty$ cells.

Previous argument relied on three Gaussian tools, and each needs a replacement.

Three Replacements

1. **Exact Gaussian moment calculations** motivated the debiasing matrix Γ_j .
 - Replaced by Lindeberg-type bounds. One-by-one, the observations making up each ξ_ℓ are replaced with Gaussian analogs of matching covariance.
2. **Stein's identity** made the SURE criterion unbiased for aggregate risk.
 - Replaced by an approximate Stein identity also proved via leave-one-observation-out argument, similar to CLT proofs via Stein's method.
3. **Gaussian Poincaré** gave concentration of SURE around realized loss.
 - Replaced by Efron–Stein argument applied at the level of individual observations.

Lindeberg

Approximate Stein

Efron–Stein

General Case: Tuning Parameter

Weights compare estimates roughly through $\gamma \|\hat{\beta}_j - \hat{\beta}_k\|^2 / \sigma^2$. Within a group this is $\gamma \|\xi_j - \xi_k\|^2$, with ξ_j, ξ_k mean-zero noise vectors in $\mathbb{R}^d$.

Problem: $\|\xi_j - \xi_k\|$ diverges as $d \rightarrow \infty$, so if γ is fixed $\omega_{jk} \approx \mathbb{1}\{k = j\}$ and the estimator reduces to the naive one.

Solution: Set $\gamma = \kappa/d$ for fixed $\kappa > 0$, so that $\gamma \|\xi_j - \xi_k\|^2 = O_p(1)$;

- Within a group, weights remain random;
- Across groups, ω_{jk} still tends to zero quickly, since the separation term grows with $1/\sigma^2$.

Proposed default $\gamma = 0.2/[d \max_{\ell} \lambda_{\max}(\Sigma_{\ell})]$ enforces this scaling.

General Case: Conditions

Roughly, the results in this section require the following:

1. **Sub-Gaussian errors.** Independent and mean zero, with uniformly bounded sub-Gaussian norms. Covers bounded errors and Gaussian tails, and allows arbitrary heteroskedasticity.
2. **Balanced cells.** Each cell holds order $n/\mathcal{J}$ observations with a well-conditioned design, and no single observation dominates its own cell.
3. **Rates and tuning.** Total number of coefficients is small relative to the sample size, and the tuning parameter scales as $\gamma = \kappa/d$.
4. **Well-separated groups.** As before, with separation now measured per coordinate. Needed only for adaptivity.
5. **Small approximation error.** Within-cell series error vanishes faster than the sampling noise.

Data Generating Process

Separation

Approximation Error

General Case: Adaptivity

Theorem 3 (Adaptivity to Exact Group Structure)

Suppose the data generating process $\mathcal{D}$, separation $\mathcal{S}$, and approximation error $\mathcal{A}$ conditions hold, with the covariate dimension growing and n large relative to Jd^2 up to logarithmic factors $\mathcal{L}$. Then

$$\sum_{j=1}^J \|\check{\beta}_j - \beta_j\|^2 = O_p \left(d \left\{ \underbrace{\frac{J}{n}}_{\text{oracle}} + \underbrace{\frac{J^3}{n^2} \log^5(Jd)}_{\text{feasibility}} \right\} \right).$$

Interpretation:

- First term is the oracle rate. An oracle pools $\asymp J$ cells per group and estimates $Gd \asymp d$ coefficients from all n observations, so its loss accumulated over the J cells is of order dJ/n .
- Second is the cost of using estimated covariances and non-Gaussian errors, negligible once $J^2 \log^5(Jd)/n \rightarrow 0$.
- Naive loss is $\asymp dJ^2/n$, so the improvement is still substantial. Loss per coefficient is $O_p \left(\frac{dJ}{n} \right)$ for naive and $O_p \left(d \left\{ \frac{1}{n} + \frac{J}{n} \right\}^2 \right)$ for proposed.

General Case: Weak Domination

Theorem 4 (Weak Domination of Least Squares)

Suppose the data generating process $\mathcal{D}$ and approximation error $\mathcal{E}$ conditions hold, with the covariate dimension growing, J large relative to $\log^{12}(Jd)$, and n large relative to Jd^2 up to slightly larger logarithmic factors $\mathcal{E}$. Then

$$\sum_{j=1}^J \|\check{\beta}_j - \beta_j\|^2 \leq (1 + o_p(1)) \sum_{j=1}^J \|\hat{\beta}_j - \beta_j\|^2.$$

Discussion:

- As before, no separation, finite-group, or exact-equality condition on $\beta_1, \dots, \beta_J$.
- Extra condition $\log^{12}(Jd)/J \rightarrow 0$, not needed for adaptivity, is mild and holds whenever J grows polynomially in n .

Rates in Both Settings

aggregate loss	
Naive	$\asymp_p d\mathfrak{J}^2/n$
Group oracle	$\asymp d\mathfrak{J}/n$
Sequence model	
General model	$O_p(d\mathfrak{J}/n + d\mathfrak{J}^3 \log^5(\mathfrak{J}d)/n^2)$

Interpretation:

- Both settings improve on the naive rate by up to a factor $\mathfrak{J}$, and both attain the oracle rate up to the terms shown;
- Second term in the general row comes from estimating the covariances without normality; the sequence model, with Σ_j known, has no such term.

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Design

Scalar cell-means design generates the sequence model exactly. Set $X_i = 1$, so β_j is the mean in cell j ;

$$Y_{ij} = \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, 1)$$

independently across observations and cells, with $J = 160$ cells and $m = n/J \in \{20, 40, 80, 160\}$.

Signal structures

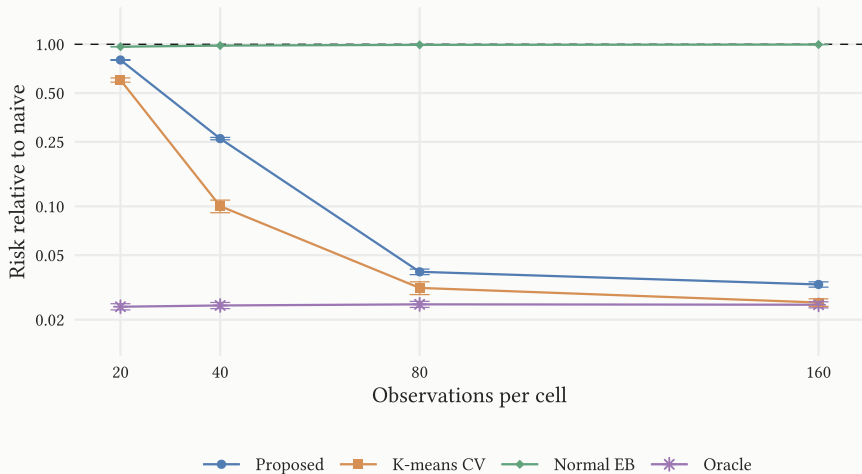
- Strong groups: $\beta_j \in \{-1.5, -0.5, 0.5, 1.5\}$;
- Weak groups: $\beta_j \in \{-0.3, -0.1, 0.1, 0.3\}$;
- Smooth: evenly spaced on $[-1.5, 1.5]$;
- Idiosyncratic: one fixed draw from $U[-1.5, 1.5]$.

Benchmarks

- Naive cell means;
 - Cross-validated k -means;
 - Normal EB (Kane et al., 2008, Abaluck et al., 2021);
 - Infeasible group oracle.
-
- Loss is aggregate squared error divided by that of the naive estimator in the same draw, so naive is one by construction; figures report the mean over 1,000 replications;
 - To match the applications, within-cell variance estimates replace the known $\Sigma_j = 1$ in the proposed and EB estimators; at the population covariance the unified default gives $\gamma = 0.2$ since $d = 1$.

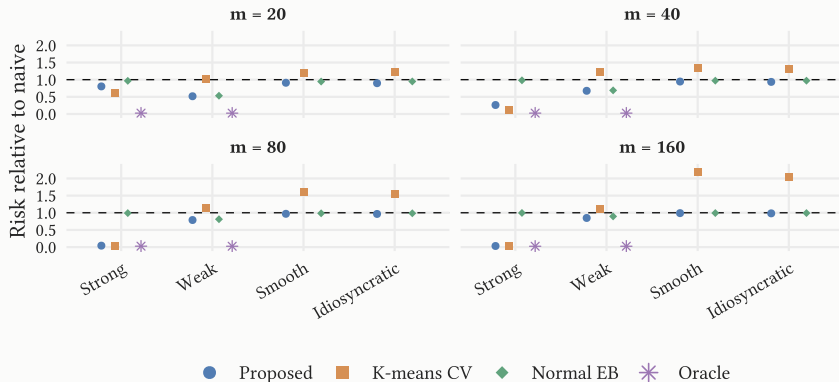
Monte Carlo: Adaptivity

Adaptation under strong group structure (Gaussian Regime)



Performance Across Designs

Performance across designs and sample sizes (Gaussian Regime)



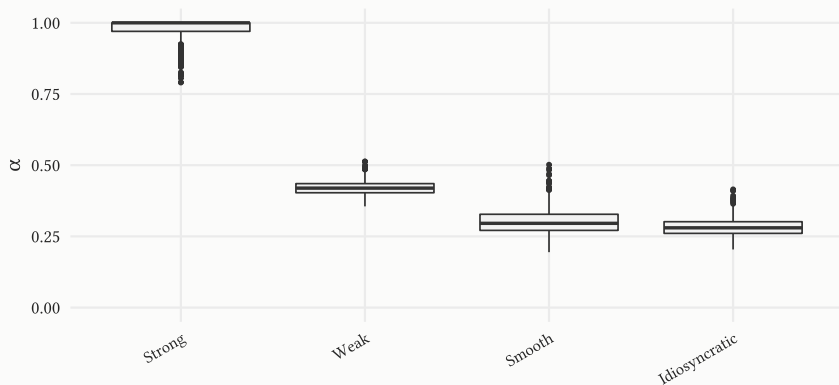
Performance Across Designs: Discussion

1. Under strong groups the estimator approaches the grouped benchmarks as noise falls. At $m = 80$ it is at 0.042, against 0.033 for k -means and 0.027 for the oracle.
 - Normal EB stays at 0.99: a single Gaussian prior fits the four-point distribution of means poorly, so the estimated prior variance is large and shrinkage is small.
 2. When groups are weak or absent it stays at or below the naive benchmark.
 - Weak groups at $m = 80$: 0.79, similar to Normal EB, while k -means is at 1.14;
 - Both no-group designs: about 0.97; k -means rises from roughly 1.6 ($m = 80$) to above 2 ($m = 160$).
- First pattern illustrates **adaptivity** ↗ and the second **weak domination** ↗;
 - Bias from an incorrect hard partition becomes more visible as sampling noise falls, SURE pulls the proposed estimator back toward the naive estimate.

Replication-Level Loss

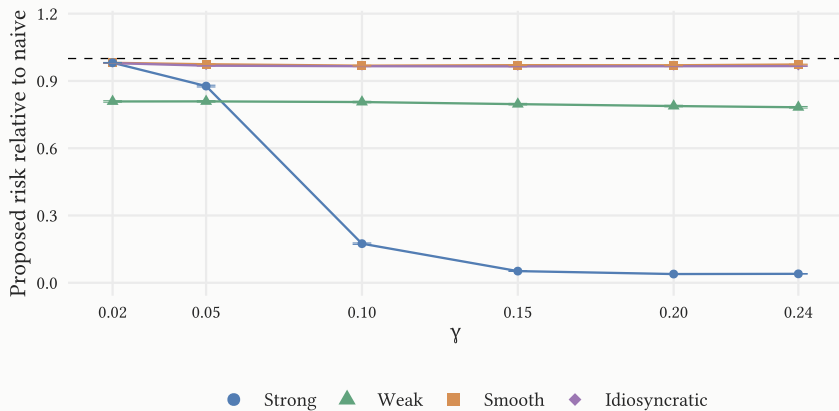
Mixing Weights

SURE mixing weights at $m = 80$ (Gaussian Regime)



Monte Carlo: Tuning Parameter

Sensitivity to gamma (Gaussian Regime)



Non-Gaussian: Design

Repeat the same exercises in a multivariate non-Gaussian design with estimated covariance matrices. Take $\beta_j \in \mathbb{R}^5$ and

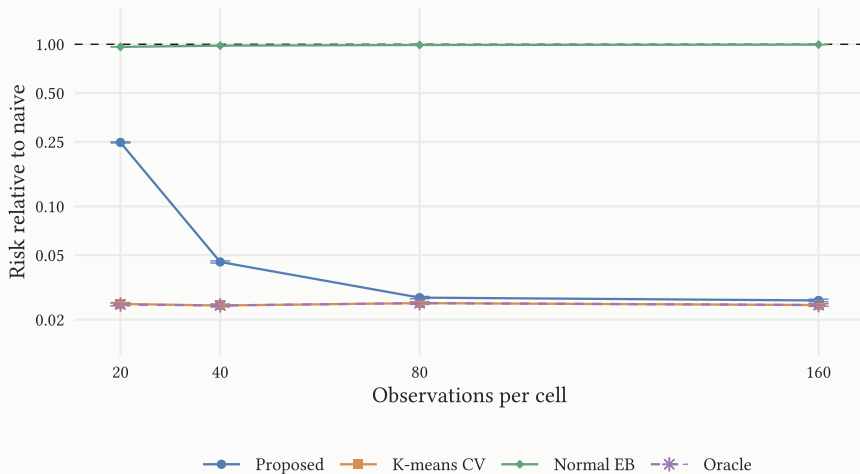
$$Y_{ij} = \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \text{ with iid coordinates drawn from } \text{Uniform}[-\sqrt{3}, \sqrt{3}]$$

so each coordinate has mean zero and unit variance but the errors are not Gaussian.

- Group centers are scalar multiples of a fixed direction vector with entries between 0.7 and 1.3; weak-group multipliers are $\{-0.09, -0.03, 0.03, 0.09\}$, so spacing is fixed per coordinate and Euclidean separation is of order $\sqrt{d}$;
- Loss is aggregate squared error, summed over cells and coordinates, again divided by that of the naive estimator in the same draw;
- Using the population covariance, $d = 5$ and $\Sigma_\ell = I_5$, so the unified default gives $\gamma = 0.2/5 = 0.04$.

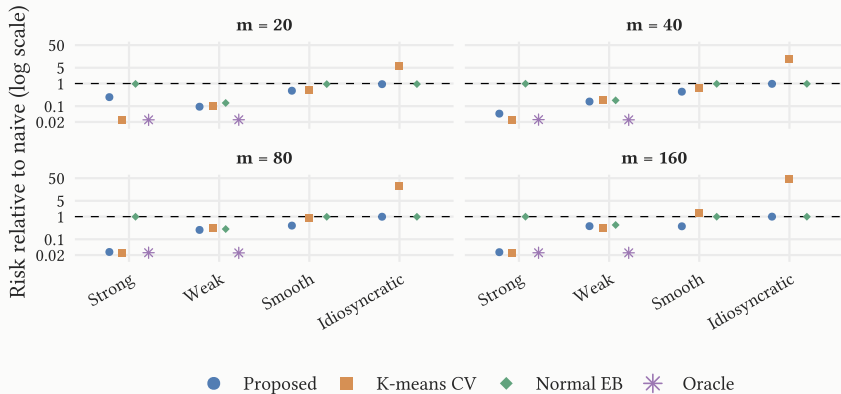
Non-Gaussian: Adaptivity

Adaptation under strong group structure (Non-Gaussian Regime)



Non-Gaussian: Performance

Performance across designs and sample sizes (Non-Gaussian Regime)



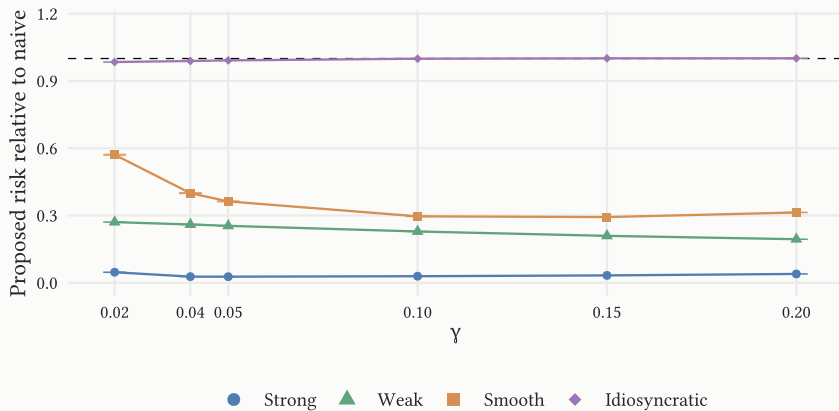
Replication-Level Loss

Mixing Weights

Runtime

Non-Gaussian: Tuning Parameter

Sensitivity to gamma (Non-Gaussian Regime)



Computation

Timing the estimator call only, taking the cell-level summaries as given.

	Proposed	k -means CV
Scalar Gaussian, $m = 80$	≈ 4 ms	13.4–15.8 ms
Multivariate uniform, $m = 80$	≈ 2.9 ms	20.7–42.4 ms

- Cost of k -means is largest in the idiosyncratic design, where the validation problem has no stable partition to select, rather than in the clearly grouped one;
- Cross-validation repeatedly solves random-start clustering problems over candidate values of G , then refits on the full cell sample. The proposed estimator uses a single deterministic weighting calculation.

Runtime

Non-Gaussian Runtime

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Firm-Level Contact Gaps

Kline et al. (2022) send more than 83,000 applications to $J = 108$ large U.S. employers, comparing contact rates for distinctively white-sounding and Black-sounding names.

Let Y_i indicate contact and W_i index the firm;

$$Y_i = \mu(W_i) + \tau(W_i)\text{WhiteName}_i + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid W_i, \text{WhiteName}_i] = 0$$

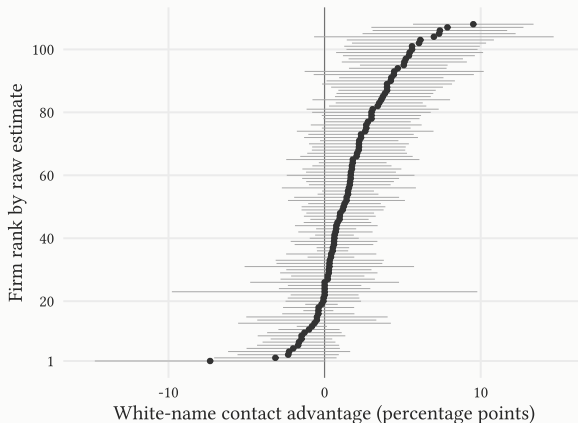
so $X_i = (1, \text{WhiteName}_i)'$ and $\beta(j) = (\mu(j), \tau(j))'$.

- Target is τ_j , firm j 's contact advantage for white-sounding names;
- We apply the scalar ($d = 1$) estimator to the published $\hat{\tau}_j$ and their standard errors;
- Group structure may arise from shared hiring practices: 83 of the 108 firms used an outsourced hiring intermediary.

Raw Estimates

Raw firm-level racial contact gaps

Dots are firm-specific estimates; horizontal bars are 95 percent intervals.

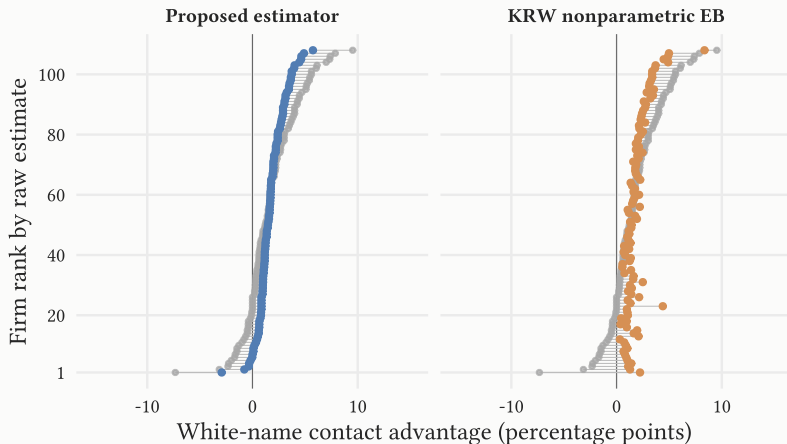


Estimates range from -7.3 to 9.5 percentage points, with a median standard error of 1.6 points; 85 of the 108 intervals cover zero.

Proposed and Empirical Bayes

Proposed and KRW nonparametric-EB racial contact gaps

Gray dots are raw estimates; colored dots are shrunk estimates.



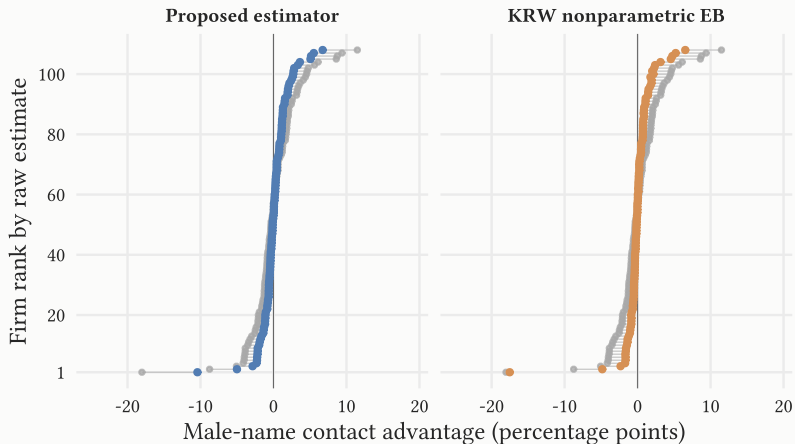
Distributions

By Industry

Gender Contact Gaps

Proposed and KRW nonparametric-EB gender contact gaps

Gray dots are raw estimates; colored dots are shrunk estimates.



First-Stage Regularization

Blandhol et al. (2026) reanalyze Card (1995), who instrument for years of schooling with proximity to a four-year college.

- **Requirement:** For 2SLS to recover a LATE, the first stage must be nonparametric in the covariates;
- **Covariate Cells:** Race, urban residence, Southern residence, 1966 urban residence, and 1966 region give $J = 92$ for $n = 3,010$, so $n/J \approx 33$;
- **Estimator:** Proposed estimator applied to the 65 cells in which proximity varies, the other 27 keep their cell mean. SURE selects $\check{\alpha} = 0.33$.

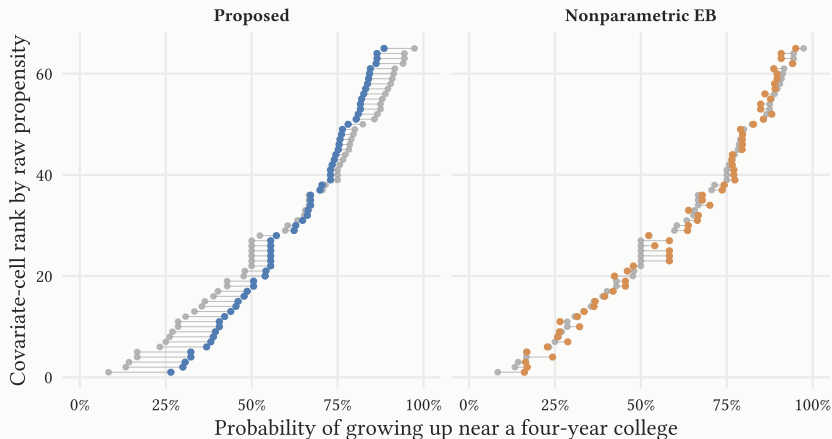
Compare to NPMLE from Jiang and Zhang (2009).

Coarsening

First-Stage Propensities

Card first-stage propensities

Gray dots are saturated cell means; colored dots are shrunk estimates.



The 27 cells in which proximity does not vary keep their cell mean and are omitted.

Distribution

Cell-Level Estimates

Return to Schooling

Estimator	Estimate	Std. error
OLS with controls	0.075	(0.004)
IV, no controls	0.188	(0.026)
Original IV with controls	0.132	(0.054)
DDML rich-covariate IV	0.122	(0.053)
Saturated first-stage	0.128	(0.055)
NPMLE first-stage	0.129	(0.057)
Proposed first-stage	0.124	(0.055)

- First four rows are benchmarks reported by Blandhol et al. (2026);
- Last three use saturated regression, NPMLE, or our proposed estimator for the first stage.

SURE keeps substantial pooling, $\check{\alpha} = 0.33$, so the cell propensities change, while the structural estimate barely moves.

Conclusion

This paper proposes an adaptive estimator for cross-sectional models in which coefficients vary with a discrete covariate of large support. This estimator

1. Pools cells that are close in a variance-adjusted metric, debiases, and recombines with the naive estimator using SURE, without estimating a partition;
2. Attains the oracle pooling rate up to logarithmic factors under well-separated groups, and asymptotically has aggregate loss no larger than the naive estimator when groups are weak or absent;
3. Extends to estimated covariances, non-Gaussian errors, growing covariate dimension, and approximation error;
4. Is shown to perform well in simulations, with or without normality, and is applied to Kline et al. (2022) and Card (1995).

R and Stata packages at <https://github.com/mnavjeev/ewgroup> and https://github.com/mnavjeev/ewgroup_stata.

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Thank you all very much

References

-  Abaluck, Jason et al. (2021). “Mortality Effects and Choice Across Private Health Insurance Plans”. In: *The Quarterly Journal of Economics* 136.3, pp. 1557–1610. DOI: 10.1093/qje/qjab017. URL: <https://doi.org/10.1093/qje/qjab017>.
-  Aloise, Daniel et al. (2009). “NP-Hardness of Euclidean Sum-of-Squares Clustering”. In: *Machine Learning* 75.2, pp. 245–248. DOI: 10.1007/s10994-009-5103-0.
-  Ando, Tomohiro and Jushan Bai (2016). “Panel Data Models with Grouped Factor Structure Under Unknown Group Membership”. In: *Journal of Applied Econometrics* 31.1, pp. 163–191. DOI: 10.1002/jae.2467.
-  Armstrong, Timothy B., Michal Kolesár, and Mikkel Plagborg-Møller (2022). “Robust Empirical Bayes Confidence Intervals”. In: *Econometrica* 90.6, pp. 2567–2602. DOI: 10.3982/ECTA18597. URL: <https://doi.org/10.3982/ECTA18597>.
-  Autor, David H. and David Dorn (2013). “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market”. In: *American Economic Review* 103.5, pp. 1553–1597. DOI: 10.1257/aer.103.5.1553. URL: <https://www.aeaweb.org/articles?id=10.1257/aer.103.5.1553>.
-  Blandhol, Christine et al. (2026). “When Is TSLS Actually LATE?” NBER Working Paper No. 29709, revised January 2025; forthcoming, *Review of Economic Studies*. DOI: 10.3386/w29709. URL: <https://www.nber.org/papers/w29709>.

References



Bonhomme, Stéphane, Thibaut Lamadon, and Elena Manresa (2022). “Discretizing Unobserved Heterogeneity”. In: *Econometrica* 90.2, pp. 625–643. DOI: <https://doi.org/10.3982/ECTA15238>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA15238>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA15238>.



Bonhomme, Stéphane and Elena Manresa (2015). “Grouped Patterns of Heterogeneity in Panel Data”. In: *Econometrica* 83.3, pp. 1147–1184. DOI: [10.3982/ECTA11319](https://doi.org/10.3982/ECTA11319).



Card, David (1995). “Using Geographic Variation in College Proximity to Estimate the Return to Schooling”. In: *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*. Ed. by Louis N. Christofides, E. Kenneth Grant, and Robert Swidinsky. Toronto: University of Toronto Press, pp. 201–222.



Chen, Jiafeng (2026). “Empirical Bayes When Estimation Precision Predicts Parameters”. In: *Econometrica* 94.2, pp. 305–340. DOI: [10.3982/ECTA22935](https://doi.org/10.3982/ECTA22935). URL: <https://doi.org/10.3982/ECTA22935>.



Chen, Xiaohong (2007). “Large Sample Sieve Estimation of Semi-Nonparametric Models”. In: *Handbook of Econometrics*. Ed. by J.J. Heckman and E.E. Leamer. 1st ed. Vol. 6B. Elsevier. Chap. 76, pp. 5549–5632. URL: <https://EconPapers.repec.org/RePEc:eee:ecochp:6b-76>.



Chetty, Raj, John N. Friedman, and Emmanuel Saez (2013). “Using Differences in Knowledge across Neighborhoods to Uncover the Impacts of the EITC on Earnings”. In: *American Economic Review* 103.7, pp. 2683–2721. DOI: [10.1257/aer.103.7.2683](https://doi.org/10.1257/aer.103.7.2683). URL: <https://www.aeaweb.org/articles?id=10.1257/aer.103.7.2683>.

References



Chetty, Raj and Nathaniel Hendren (2018). “The Impacts of Neighborhoods on Intergenerational Mobility II: County-Level Estimates”. In: *The Quarterly Journal of Economics* 133.3, pp. 1163–1228. DOI: 10.1093/qje/qjy006. URL: <https://doi.org/10.1093/qje/qjy006>.



Chetverikov, Denis and Elena Manresa (forthcoming). “Spectral and Post-Spectral Estimators for Grouped Panel Data Models”. In: *Review of Economic Studies*. Forthcoming. URL: <https://denischetverikov.wordpress.com/wp-content/uploads/2026/04/spectralestimator.v27.main.text.pdf>.



Dahl, Gordon B., Andreas Ravndal Kostøl, and Magne Mogstad (2014). “Family Welfare Cultures”. In: *The Quarterly Journal of Economics* 129.4, pp. 1711–1752. DOI: 10.1093/qje/qju019. URL: <https://doi.org/10.1093/qje/qju019>.



Dasgupta, Sanjoy (2008). *The Hardness of k-Means Clustering*. Tech. rep. CS2008-0916. Department of Computer Science and Engineering, University of California, San Diego. URL: <https://cseweb.ucsd.edu/~dasgupta/papers/kmeans.pdf>.



Durand, Frédo and Julie Dorsey (2002). “Fast Bilateral Filtering for the Display of High-Dynamic-Range Images”. In: *ACM Transactions on Graphics* 21.3, pp. 257–266. DOI: 10.1145/566570.566574.



Gu, Jiaying and Roger Koenker (2023). “Invidious Comparisons: Ranking and Selection as Compound Decisions”. In: *Econometrica* 91.1, pp. 1–41. DOI: 10.3982/ECTA19304. URL: <https://doi.org/10.3982/ECTA19304>.

References

-  James, W. and Charles Stein (1961). “Estimation with Quadratic Loss”. In: *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*. Berkeley, CA: University of California Press, pp. 361–379. URL: <https://projecteuclid.org/euclid.bsmmsp/1200512173>.
-  Jiang, Wenhua and Cun-Hui Zhang (2009). “General maximum likelihood empirical Bayes estimation of normal means”. In: *The Annals of Statistics* 37.4, pp. 1647–1684.
-  Kane, Thomas J., Jonah E. Rockoff, and Douglas O. Staiger (2008). “What Does Certification Tell Us About Teacher Effectiveness? Evidence from New York City”. In: *Economics of Education Review* 27.6, pp. 615–631. DOI: 10.1016/j.econedurev.2007.05.005. URL: <https://doi.org/10.1016/j.econedurev.2007.05.005>.
-  Kline, Patrick, Evan K. Rose, and Christopher R. Walters (2022). “Systemic Discrimination Among Large U.S. Employers”. In: *The Quarterly Journal of Economics* 137.4, pp. 1963–2036. DOI: 10.1093/qje/qjac024. URL: <https://doi.org/10.1093/qje/qjac024>.
-  Lin, Chang-Ching and Serena Ng (2012). “Estimation of Panel Data Models with Parameter Heterogeneity When Group Membership is Unknown”. In: *Journal of Econometric Methods* 1.1, pp. 42–55. DOI: 10.1515/2156-6674.1000.
-  Mahajan, Meena, Prajakta Nimbhorkar, and Kasturi Varadarajan (2012). “The Planar k-Means Problem is NP-Hard”. In: *Theoretical Computer Science* 442, pp. 13–21. DOI: 10.1016/j.tcs.2010.05.034.

References



Paris, Sylvain and Frédo Durand (2006). “A Fast Approximation of the Bilateral Filter Using a Signal Processing Approach”. In: *Computer Vision – ECCV 2006*, pp. 568–580. doi: 10.1007/11744085_44.



Paris, Sylvain et al. (2009). “Bilateral Filtering: Theory and Applications”. In: *Foundations and Trends in Computer Graphics and Vision* 4.1, pp. 1–73. doi: 10.1561/06000000020.



Polyanskiy, Yury and Yihong Wu (2021). “Sharp regret bounds for empirical Bayes and compound decision problems”. arXiv:2109.03943. URL: <https://arxiv.org/abs/2109.03943>.



Robbins, Herbert (1956). “An Empirical Bayes Approach to Statistics”. In: *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*. Berkeley, CA: University of California Press, pp. 157–164. doi: 10.1525/9780520313880-015. URL: <https://doi.org/10.1525/9780520313880-015>.



Saha, Sujayam and Adityanand Guntuboyina (2020). “On the nonparametric maximum likelihood estimator for Gaussian location mixture densities with application to Gaussian denoising”. In: *The Annals of Statistics* 48.2, pp. 738–762. doi: 10.1214/19-AOS1817. URL: <https://doi.org/10.1214/19-AOS1817>.



Soloff, Jake A., Adityanand Guntuboyina, and Bodhisattva Sen (2025). “Multivariate, heteroscedastic empirical Bayes via nonparametric maximum likelihood”. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 87.1, pp. 1–32. doi: 10.1093/jrsssb/qkae040. URL: <https://doi.org/10.1093/jrsssb/qkae040>.

References

-  Song, Myunghyun, Sokbae Lee, and Serena Ng (2026). “Empirical Bayes estimation in heterogeneous coefficient panel models”. arXiv:2601.07059. URL: <https://arxiv.org/abs/2601.07059>.
-  Stein, Charles M. (1981). “Estimation of the Mean of a Multivariate Normal Distribution”. In: *The Annals of Statistics* 9.6, pp. 1135–1151. DOI: 10.1214/aos/1176345632. URL: <https://doi.org/10.1214/aos/1176345632>.
-  Stone, Charles J. (1985). “Additive Regression and Other Nonparametric Models”. In: *The Annals of Statistics* 13.2, pp. 689–705. DOI: 10.1214/aos/1176349548. URL: <https://doi.org/10.1214/aos/1176349548>.
-  Su, Liangjun, Zhentao Shi, and Peter C. B. Phillips (2016). “Identifying Latent Structures in Panel Data”. In: *Econometrica* 84.6, pp. 2215–2264. DOI: 10.3982/ECTA12560.
-  Tomasi, Carlo and Roberto Manduchi (1998). “Bilateral Filtering for Gray and Color Images”. In: *Proceedings of the Sixth International Conference on Computer Vision*, pp. 839–846. DOI: 10.1109/ICCV.1998.710815.
-  Walters, Christopher R. (2024). “Empirical Bayes Methods in Labor Economics”. In: *Handbook of Labor Economics*. Vol. 5. Elsevier. Chap. 3, pp. 183–260. DOI: 10.1016/bs.heslab.2024.11.001. URL: <https://doi.org/10.1016/bs.heslab.2024.11.001>.
-  Yu, Lu, Jiaying Gu, and Stanislav Volgushev (2024). “Spectral Clustering with Variance Information for Group Structure Estimation in Panel Data”. In: *Journal of Econometrics* 241.1, p. 105709. DOI: 10.1016/j.jeconom.2024.105709.

References



Zhang, Yan and Stanislav Volgushev (2026). “Adaptivity of the NPMLE to finitely discrete mixing distributions in Gaussian/Poisson mixtures”. arXiv:2604.12087. URL: <https://arxiv.org/abs/2604.12087>.

Appendix: Technical Details

Formal Statement: Gaussian Sequence Model

Assumption (Gaussian Sequence Model)

As $n, \mathfrak{J} \rightarrow \infty$, suppose that

- (i) d is fixed and $\hat{\beta}_j = \beta_j + \sigma \xi_j$, $\xi_j \sim N(0, \Sigma_j)$, independently across j , with $\sigma^2 = \mathfrak{J}/n \rightarrow 0$;
- (ii) The known, nonrandom matrices Σ_j are symmetric and, uniformly over j ,
 $\underline{\lambda} \leq \lambda_{\min}(\Sigma_j) \leq \lambda_{\max}(\Sigma_j) \leq \bar{\lambda}$;
- (iii) γ is fixed and satisfies $0 < \gamma < (p\bar{\lambda})^{-1}$ for some $p > 4$.

Formal Statement: Well-Separated Parameters

Assumption (Well-Separated Parameters)

There are fixed constants $\delta > 0$, $\tilde{B} < \infty$, and $G < \infty$ such that

- (i) If $\beta_k \neq \beta_\ell$ then $\|\beta_k - \beta_\ell\| \geq \delta$, and $\max_{j \in [J]} \|\beta_j\| \leq \tilde{B}$;
- (ii) The number of distinct values in $\{\beta_1, \dots, \beta_J\}$ is at most G ;
- (iii) For $g(j) = \{k \in [J] : \beta_k = \beta_j\}$, $|g(j)| \asymp J$ uniformly over j .

Imposed only for the adaptivity result.

Formal Statement: Data Generating Process

Write

$$\mathcal{I}_j := \{i : W_i = j\}, \quad G_j := \sum_{i \in \mathcal{I}_j} X_i X_i', \quad S_j := \sum_{i \in \mathcal{I}_j} \tau_i^2 X_i X_i'$$

Assumption (Data Generating Process)

- (i) Errors ε_i are independent with $\mathbb{E}[\varepsilon_i] = 0$, $\text{Var}(\varepsilon_i) = \tau_i^2$, $\underline{\tau} \leq \min_i \tau_i \leq \max_i \tau_i \leq \bar{\tau}$, and $\max_{i \leq n} \|\varepsilon_i / \tau_i\|_{\psi_2} \leq K_\varepsilon$;
 - (ii) Cells are balanced, $n_j \asymp n/\mathcal{J}$, with the scaled Gram matrix $n_j^{-1} G_j$, the scaled variance matrix $n_j^{-1} S_j$, and $\Sigma_j = \sigma^{-2} \text{Var}(\hat{\beta}_j)$ all having eigenvalues bounded above and below;
 - (iii) Maximum within-group leverage satisfies $h_n := \max_j \max_{i \in \mathcal{I}_j} X_i' G_j^{-1} X_i \lesssim d\mathcal{J}/n$;
 - (iv) The tuning parameter satisfies $\gamma = \kappa/d$ for a fixed $\kappa > 0$;
 - (v) $\mathcal{J} \rightarrow \infty$, $d\mathcal{J}/n \rightarrow 0$, and $\log(\mathcal{J}d) \lesssim d$.
-
- Part (i) replaces Gaussianity with sub-Gaussian, heteroskedastic errors;
 - Part (v) keeps the total number of coefficients $d\mathcal{J}$ small relative to n .

Formal Statement: Well-Separated Parameters (Growing d)

Assumption (Well-Separated Parameters)

There are fixed constants $\delta > 0$, $\bar{B} < \infty$, and $G < \infty$ such that

- (i) $\min_{k, \ell: \beta_k \neq \beta_\ell} \|\beta_k - \beta_\ell\| \geq \sqrt{d} \delta$ and $\max_{j \in [J]} \|\beta_j\| \leq \sqrt{d} \bar{B}$;
- (ii) The number of distinct values in $\{\beta_1, \dots, \beta_J\}$ is at most G ;
- (iii) $|g(j)| \asymp J$ uniformly over j .

Formal Statement: Approximation Error

Assumption (Approximation Error)

For $\bar{r}_{2,j}^2 := n_j^{-1} \sum_{i \in I_j} r_i^2$, the within-cell L_2 approximation error satisfies

$$\log^4(Jd) \max_{j \leq J} \bar{r}_{2,j} = o(n^{-1/2} + J/n).$$

Root mean square approximation error within each cell vanishes faster than the parametric noise scale $n^{-1/2} + J/n$, up to a logarithmic factor, so bias of the preliminary estimates and of the variance estimates is small relative to sampling noise.

Formal Statement: Rate Conditions

Both general-case results take $d \rightarrow \infty$. In addition,

- Theorem 3 (adaptivity) requires

$$\frac{\mathfrak{J}d^2 \log^4(\mathfrak{J}d)}{n} \rightarrow 0;$$

- Theorem 4 (weak domination) requires

$$\frac{\log^{12}(\mathfrak{J}d)}{\mathfrak{J}} + \frac{\mathfrak{J}d^2 \log^6(\mathfrak{J}d)}{n} \rightarrow 0.$$

 Adaptivity

 Weak Domination

Closed-Form Jacobian

Derivative entering the SURE criterion holds the preliminary (naive) estimates fixed, and in the feasible version, hold the covariance estimates fixed as well. Write

$$H_j := (I_d - \Gamma_j)^{-1}, \quad N_j := \sum_{\ell=1}^J \omega_{j\ell} (\hat{\beta}_\ell - \hat{\beta}_j), \quad \tilde{\beta}_j = \hat{\beta}_j + H_j N_j,$$

so $\hat{\beta}_j$ enters directly through N_j and indirectly through the weights. Gradient of the log-weight in $\hat{\beta}_j$ is $g_{j\ell} := \sigma^{-2} \Omega_\ell (\hat{\beta}_\ell - \hat{\beta}_j)$; centering it, $c_{j\ell} := g_{j\ell} - \sum_{k=1}^J \omega_{jk} g_{jk}$, gives $\partial \omega_{j\ell} / \partial \hat{\beta}_j = \omega_{j\ell} c'_{j\ell}$. Then

$$P_j := \frac{\partial \tilde{\beta}_j}{\partial \hat{\beta}_j} = \underbrace{I_d - (1 - \omega_{jj})H_j}_{\text{weights held fixed}} + H_j \underbrace{\sum_{\ell=1}^J \omega_{j\ell} (\hat{\beta}_\ell - \hat{\beta}_j + \gamma \Sigma_\ell H_j N_j)}_{\text{weights vary}} c'_{j\ell}$$

Every term is already computed for $\tilde{\beta}_j$, so no numerical differentiation is needed.

Stein's Unbiased Risk Estimate

Lemma (Stein's Identity)

Let $\xi \sim N(0, \Sigma)$ in $\mathbb{R}^d$ and let $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be almost differentiable with $\mathbb{E}\|Df(\xi)\| < \infty$, where $Df(x) := \partial f(x)/\partial x'$. Then

$$\mathbb{E}\langle f(\xi), \xi \rangle = \mathbb{E} \operatorname{tr} (\Sigma Df(\xi)).$$

Lemma (Stein's Unbiased Risk Estimate)

In the sequence model, let $\hat{\theta}_j = \hat{\theta}_j(\hat{\beta}_1, \dots, \hat{\beta}_j)$ be almost differentiable in $\hat{\beta}_j$ with $\mathbb{E}\|\partial \hat{\theta}_j / \partial \hat{\beta}_j\| < \infty$, $j = 1, \dots, J$. Then

$$\text{SURE}(\hat{\theta}) := \sum_{j=1}^J \|\hat{\beta}_j - \hat{\theta}_j\|^2 + 2\sigma^2 \sum_{j=1}^J \operatorname{tr} \left(\Sigma_j \frac{\partial \hat{\theta}_j}{\partial \hat{\beta}_j} \right) - \sigma^2 \sum_{j=1}^J \operatorname{tr}(\Sigma_j)$$

satisfies $\mathbb{E} \text{SURE}(\hat{\theta}) = \mathbb{E} \sum_{j=1}^J \|\hat{\theta}_j - \beta_j\|^2$.

Idea. Expand $\|\hat{\theta}_j - \beta_j\|^2 = \|\hat{\theta}_j - \hat{\beta}_j\|^2 + 2\sigma \langle \hat{\theta}_j - \hat{\beta}_j, \xi_j \rangle + \sigma^2 \|\xi_j\|^2$ and apply Stein's identity to ξ_j conditional on the other cells. Criterion $\hat{\mathcal{L}}(\alpha)$ is $\text{SURE}(\check{\beta}(\alpha))$.

Lindeberg Replacement

Gaussian moment calculations behind Γ_j are replaced by an interpolation bound.

Lemma (Third-Order Lindeberg)

Let $X_1, \dots, X_N, Z_1, \dots, Z_N$ be mutually independent mean-zero random vectors in $\mathbb{R}^d$ with $\text{Var}(X_i) = \text{Var}(Z_i)$ for each i . Put $S = \sum_i X_i$ and $S^Z = \sum_i Z_i$. If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is three times continuously differentiable with $M_3(f) := \sup_x \sup_{\|u\|=\|v\|=\|w\|=1} |D^3 f(x)[u, v, w]| < \infty$, then

$$|\mathbb{E}f(S) - \mathbb{E}f(S^Z)| \leq \frac{M_3(f)}{6} \sum_{i=1}^N (\mathbb{E}\|X_i\|^3 + \mathbb{E}\|Z_i\|^3).$$

Application: Used to show that certain moments of the exponentially weighted estimator in the general model are close to their moments in the Gaussian model, and thus that the exponentially weighted estimator is approximately unbiased conditional on $\hat{\beta}_j$.

Approximate Stein

Stein's identity, which makes the SURE criterion unbiased, relies on Gaussianity. Without normality an approximate version holds for smooth functions of a sum of independent terms.

Lemma (Approximate Stein Identity)

Let $S = \sum_{i=1}^N X_i$ with $X_1, \dots, X_N$ independent mean-zero random vectors in $\mathbb{R}^d$ with finite third moments, and put $V := \text{Var}(S)$. If $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is twice continuously differentiable with $M_2(f) := \sup_x \sup_{\|u\|=\|v\|=1} \|D^2 f(x)[u, v]\| < \infty$, then

$$\left| \mathbb{E} \langle f(S), S \rangle - \mathbb{E} \text{tr} (V Df(S)) \right| \leq \frac{3}{2} M_2(f) \sum_{i=1}^N \mathbb{E} \|X_i\|^3.$$

Application: Used to show that the SURE criterion is approximately unbiased for the true loss in the general model.

Efron–Stein

SURE must concentrate around realized loss uniformly over $\alpha \in [0, 1]$. In the Gaussian case this follows from the Poincaré inequality; without normality, from Efron–Stein.

Lemma (Efron–Stein)

Let $X_1, \dots, X_N, X'_1, \dots, X'_N$ be independent with X'_i distributed as X_i . For $X = (X_1, \dots, X_N)$, let $X^{(i)}$ replace X_i by X'_i . If $f(X) \in L^2$, then

$$\text{Var}(f(X)) \leq \frac{1}{2} \sum_{i=1}^N \mathbb{E}[(f(X) - f(X^{(i)}))^2].$$

If, further $\sup_x \|\nabla f(x)\|$ is bounded, then

$$\text{Var}(f(X)) \leq \sup_x \|\nabla f(x)\|_\infty^2 \frac{1}{2} \sum_{i=1}^N \mathbb{E}[|X_i - X'_i|^2] = \sup_x \|\nabla f(x)\|_\infty \sum_{i=1}^N \text{Var}(X_i).$$

Application: Applied to the SURE criterion to show that it concentrates around its expectation. Approximate Stein-inequalities show that this expectation is nearly the true loss, so SURE is consistent.

Gaussian Poincaré Inequality

Concentration of the SURE criterion around realized loss in the sequence model established using Gaussian Poincaré Inequality.

Lemma (Gaussian Poincaré Inequality)

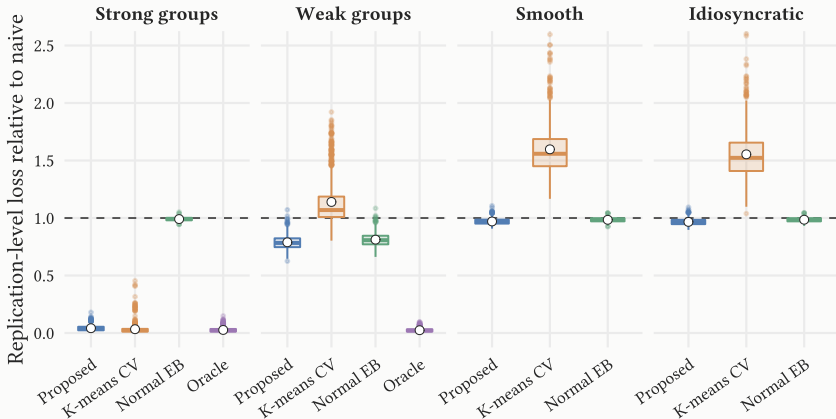
Let $\xi = (\xi_1, \dots, \xi_j)$ with $\xi_j \sim N(0, \Sigma_j)$ independent across j , and let $f : \mathbb{R}^{jd} \rightarrow \mathbb{R}$ be almost differentiable with $\mathbb{E}\|\nabla f(\xi)\|^2 < \infty$. Then

$$\text{Var}(f(\xi)) \leq \sum_{j=1}^J \mathbb{E}[\nabla_j f(\xi)' \Sigma_j \nabla_j f(\xi)] \leq \max_j \lambda_{\max}(\Sigma_j) \mathbb{E}\|\nabla f(\xi)\|^2,$$

where $\nabla_j f$ is the gradient with respect to ξ_j .

Replication-Level Loss

Distribution of relative loss across replications (Gaussian Regime)

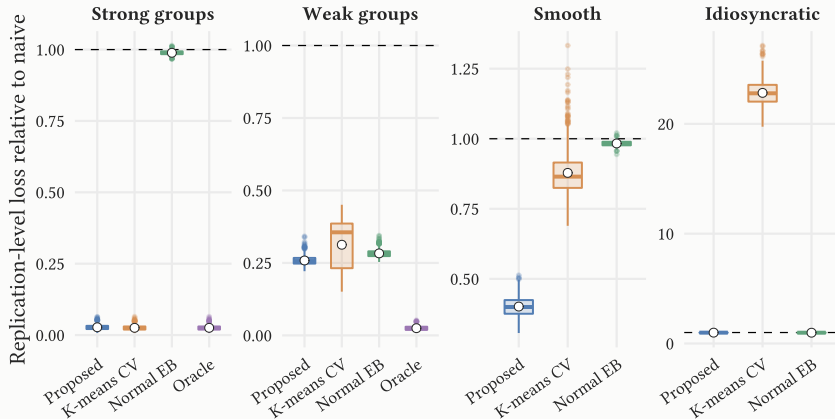


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[Non-Gaussian](#)

Non-Gaussian: Replication-Level Loss

Distribution of relative loss across replications (Non-Gaussian Regime)

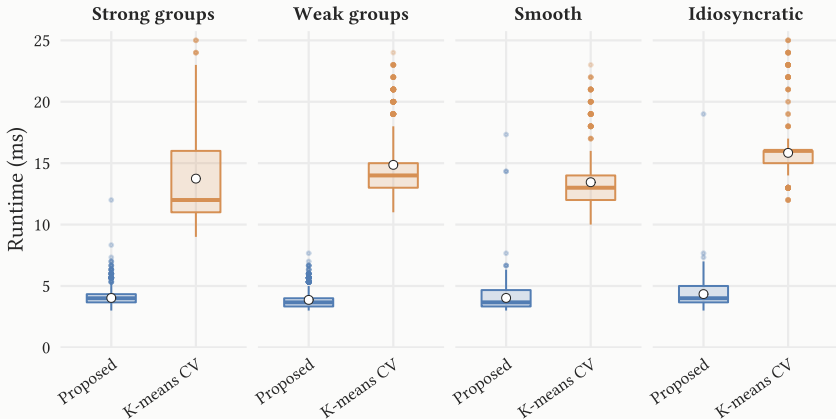


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[Non-Gaussian Performance](#)

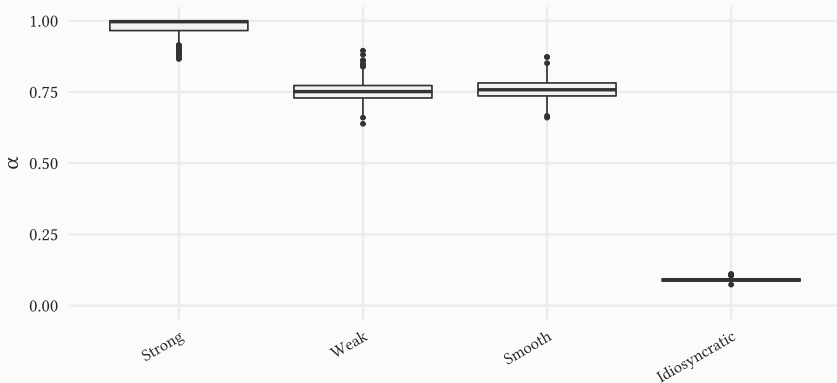
Runtime

Distribution of runtime across replications (Gaussian Regime)



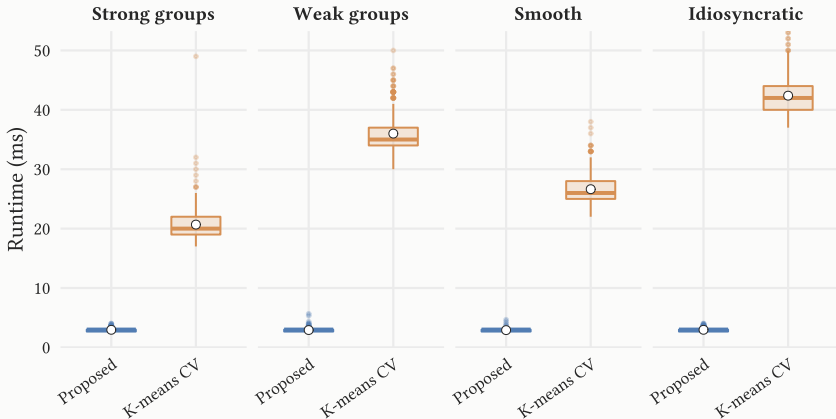
Non-Gaussian: Mixing Weights

SURE mixing weights at $m = 80$ (Non-Gaussian Regime)



Non-Gaussian: Runtime

Distribution of runtime across replications (Non-Gaussian Regime)



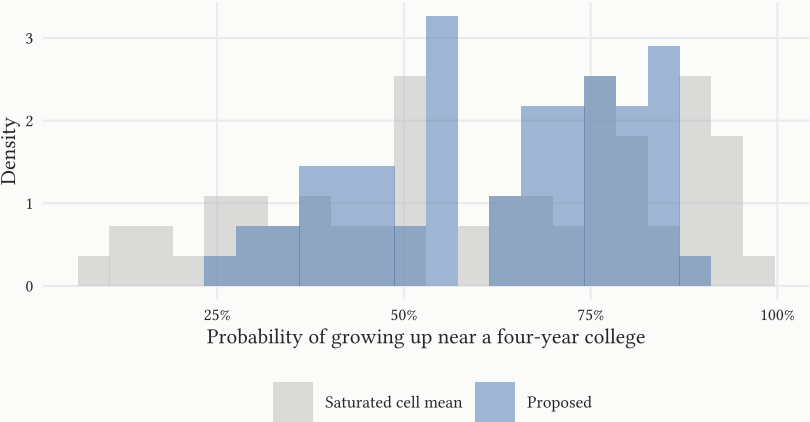
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[Computation](#)

Distribution of First-Stage Propensities

Proposed propensities run from 27 to 89 percent, against 8 to 97 percent for the cell means.

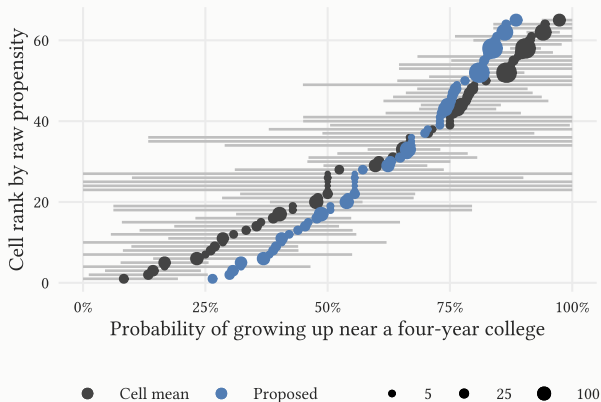
Distribution of raw and proposed first-stage propensities



Cell-Level Estimates

Cells at the low end of the ranking move most, by up to 18 percentage points, while cells with at least 100 observations move by at most 9.

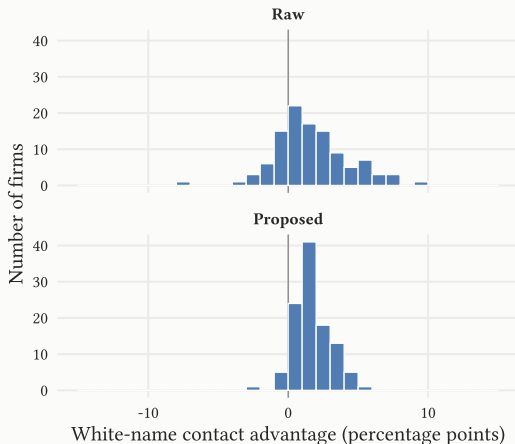
First-stage propensity estimates in the Card application



Distribution of Gaps

Pooling halves the spread of the firm estimates, from a standard deviation of 2.6 to 1.3 percentage points.

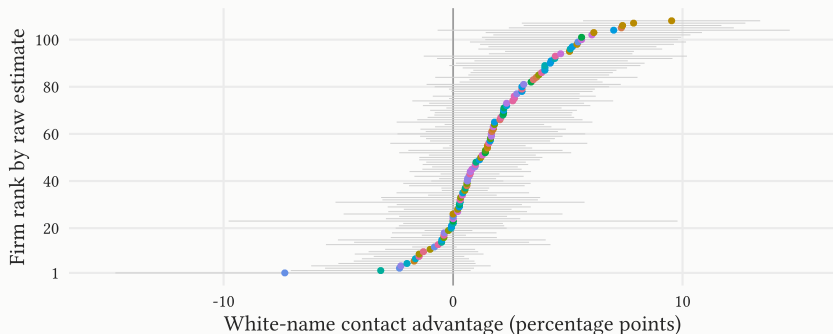
Raw and adaptive shrunk contact-gap distributions



By Industry

Raw contact gaps by industry

Colors show two-digit industry categories in the KRW replication data.



- | | | | |
|---------------------------|------------------------|-----------------------|---------------------------|
| ● Accommodation | ● Building materials | ● Food products | ● Insurance / real estate |
| ● Apparel manufacturing | ● Business services | ● Food stores | ● Other manufacturing |
| ● Apparel stores | ● Communications | ● Freight / transport | ● Other retail |
| ● Auto / repair services | ● Eating/drinking | ● Furnishing stores | ● Securities brokers |
| ● Auto dealers / services | ● Electric / gas | ● General merchandise | ● Wholesale durable |
| ● Banks / credit | ● Engineering services | ● Health services | ● Wholesale nondurable |

Coarsening the Covariate Cells

A fully saturated first stage would also interact potential experience with the discrete controls, giving $J = 819$ cells with a median of two observations each and 358 singletons, too few for theory to be applicable.

- The application coarsens to race, urban residence, Southern residence, 1966 urban residence, and 1966 region: $J = 92$ cells for $n = 3,010$, so $n/J \approx 33$;
- The raw proportion is still zero or one in 27 cells, where the plug-in variance vanishes and the eigenvalue bound on Σ_J fails;
- Those 27 cells, 125 observations in all, keep their cell mean. Estimator pools the remaining 65 cells, with a mean of 44 observations each.

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Card Table Notes

First four rows are the benchmarks reported by [Blandhol et al. \(2026\)](#), Table 3; the last three are computed from the Wooldridge/Card data.

DDML row:

- [Blandhol et al. \(2026\)](#) estimate the optimal instrument using an ensemble method where predictions are averaged between: a random forest with 1000 trees, gradient boosted trees with 1000 stages, and a neural network with two neurons method where predictions are averaged between.

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Coverage of the Scalar Applications

Both applications and the scalar Monte Carlo run the $d = 1$ estimator with estimated variances, which neither set of results covers directly.

- Sequence-model results take Σ_j known and the noise Gaussian; the general results relax both but take $d \rightarrow \infty$.
- Sequence experiment is motivated by a within-cell central limit theorem. [Kline et al. \(2022\)](#) has $n/\mathcal{J} \approx 770$; [Card \(1995\)](#) pools 65 cells with a median of 23 observations and a minimum of 2, so the approximation is less reliable there.
- Cells in [Card \(1995\)](#) without variation in proximity have zero plug-in variance; they keep their cell mean and stay out of the pooling.

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Bilateral Filtering

Step 1 is related to bilateral filtering, a method from image processing (Tomasi and Manduchi, 1998, Durand and Dorsey, 2002, Paris and Durand, 2006, Paris et al., 2009).

A bilateral filter smooths an image by averaging a pixel with others that are nearby and similar in color or brightness, so information is borrowed from similar objects while sharp differences are preserved.

Exponential weights use the same principle: preliminary cell-specific estimates that are close to one another are averaged more heavily.

- Goals differ. That literature is concerned with image processing and computation; here the weighted average is one step of an estimator with large-sample guarantees.
- Debiasing and SURE recombination are added to obtain those guarantees.